Type 2 Paper: A Critical Methodological Review of Hybrid Machine Learning Techniques in Housing Price Prediction

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Abstract
This paper critically reviews the methodologies of three different works on housing price prediction, exploring advanced hybrid machine learning models and structural economic frameworks. The first study introduces hybrid machine learning techniques, employing stacked generalization and regression models on the "Housing Price in Beijing" dataset. It emphasizes the importance of feature engineering and ensemble methods to improve predictive accuracy. The second study presents a hybrid framework that integrates Bayesian Optimization with stacking, bagging, and Transformer-based techniques, applied to a large-scale Hong Kong dataset. It evaluates 27 models using 13 metrics, demonstrating the superior performance of hybrid models in predictive tasks. The third examines the housing market's boom and bust during the Great Recession through a structural equilibrium model, analyzing the dynamics of house prices, consumption, and credit. It highlights the significant role of belief shifts over credit conditions in driving house price fluctuations. This review delves into the methodologies, highlighting key variables, equations, and evaluation strategies used across the studies. The innovations in hybrid machine learning and structural modeling significantly contribute to improving housing price predictions. However, challenges remain, including model interpretability and real-time data integration. The paper concludes by identifying future research directions to address these challenges, emphasizing the need for more dynamic, scalable, and interpretable models.

Keywords
Housing Price Prediction, Hybrid Regression, Machine Learning, Stacked Generalization, Ensemble Models, Housing Boom and Bust, Structural Equilibrium Model, Aggregate Shocks, Belief Shifts, Credit Conditions, Bayesian Optimization, Stacking and Bagging, Transformer Models

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1. Introduction

Accurately forecasting housing prices is a critical and complex task, given the non-linear interplay between economic factors, property attributes, and market dynamics. Traditional approaches, such as the House Price Index (HPI) and hedonic regression models, often fall short in capturing these complexities, particularly at the granular level of individual property pricing. To address these limitations, recent advancements have focused on hybrid machine learning techniques that leverage ensemble methods and advanced optimization strategies.

This paper reviews three key studies that exemplify the application of advanced methodologies in housing price prediction. The first study explores hybrid regression models and stacked generalization, utilizing the "Housing Price in Beijing" dataset. It emphasizes the importance of feature engineering and ensemble techniques in improving predictive performance, demonstrating a significant enhancement over traditional models. The second study takes a step further by integrating Bayesian Optimization with stacking, bagging, and Transformer-based techniques. Applied to a comprehensive dataset of Hong Kong real estate transactions, this approach underscores the role of hyperparameter optimization and sequential learning in achieving superior prediction accuracy.

The third study shifts focus to the macroeconomic factors influencing housing markets, particularly during the Great Recession. Kaplan et al. employ a structural equilibrium model to analyze the dynamics of house prices, consumption, and credit conditions. By simulating aggregate shocks and belief shifts, they reveal that shifts in expectations, rather than changes in credit constraints, played a pivotal role in the housing boom and bust cycle. Together, these studies provide a comprehensive understanding of housing price prediction, bridging the gap between data-driven machine learning models and macroeconomic insights, and offering a robust framework for future research and practical applications

2. Background, Literature Review, and Motivating Factors

2.1 Evolution of Housing Price Prediction Models

Accurately predicting housing prices has long been a challenge for economists and data scientists due to the complex, non-linear relationships between economic indicators, property attributes, and market dynamics. Early models like hedonic pricing and ARIMA served as foundational tools for housing price analysis. These traditional approaches, while useful for capturing basic linear relationships, struggled to address the intricacies of modern housing markets, especially during periods of significant market fluctuations or crises.

To overcome these limitations, the field has increasingly turned to machine learning (ML) methods. Models such as decision trees, random forests, gradient boosting, and neural networks have demonstrated the capability to capture non-linear interactions and improve predictive accuracy.

However, these methods often come with trade-offs, including the need for extensive hyperparameter tuning and high computational costs.

2.2 Hybrid Approaches in Prediction

Building on the advancements of standalone machine learning models, hybrid approaches have emerged as powerful tools for housing price prediction. Hybrid models, such as those leveraging bagging, boosting, and stacking, aim to combine the strengths of individual algorithms while mitigating their weaknesses. For instance, ensemble methods like random forests reduce variance by aggregating predictions from multiple decision trees, while boosting techniques like XGBoost focus on minimizing bias by iteratively improving weaker learners.

A significant innovation in this domain is the integration of Bayesian Optimization (BO), which automates the hyperparameter tuning process to enhance model performance. This approach has been successfully applied in hybrid models such as HBOS (Hybrid Bayesian Optimization with Stacking), HBOB (with Bagging), and HBOT (with Transformer-based architectures). These models demonstrate superior performance by optimizing complex, high-dimensional parameter spaces, particularly in large datasets like Hong Kong’s real estate transactions.

2.3 Dynamic Modeling of Housing Markets

While machine learning focuses on predictive accuracy at a micro level, macroeconomic models provide insights into the underlying drivers of housing market trends. Kaplan,

Mitman, and Violante introduce a structural equilibrium model to explore the macroeconomic dynamics of the housing market during the Great Recession. Their model incorporates overlapping generations and incomplete markets, allowing for a detailed analysis of three critical aggregate shocks: labor productivity, credit conditions, and shifts in household beliefs about future housing demand.

Unlike traditional macroeconomic models, which often emphasize credit constraints as the primary driver of housing market fluctuations, Kaplan et al.'s work highlights the pivotal role of belief shifts. Their findings suggest that changes in household expectations significantly influence house prices and consumption patterns, particularly during periods of economic instability.

2.4 Contributions of the Reviewed Studies

Each of the reviewed studies contributes uniquely to the field of housing price prediction. The first study focuses on improving predictive performance through stacked generalization and hybrid regression techniques, demonstrating their efficacy on the "Housing Price in Beijing" dataset. The second study further advances the methodology by integrating Bayesian Optimization with ensemble methods, providing a robust framework for optimizing housing price forecasts in dynamic markets. Finally, the third study complements these data-driven approaches by offering a macroeconomic perspective, using structural equilibrium modeling to uncover the fundamental drivers of housing market cycles.

Together, these studies illustrate the evolution of housing price prediction, from traditional econometric models to cutting-edge hybrid machine learning frameworks and provide a comprehensive toolkit for tackling the multifaceted challenges of modern housing markets.

3. Research Objectives

The reviewed studies aim to address specific challenges in housing price prediction:
- Paper 1: Improve prediction accuracy through hybrid regression techniques.
- Paper 2: Identify the macroeconomic factors driving housing market fluctuations.
- Paper 3: Enhance model robustness and accuracy using hybrid machine learning frameworks.

4. Methodological Analysis

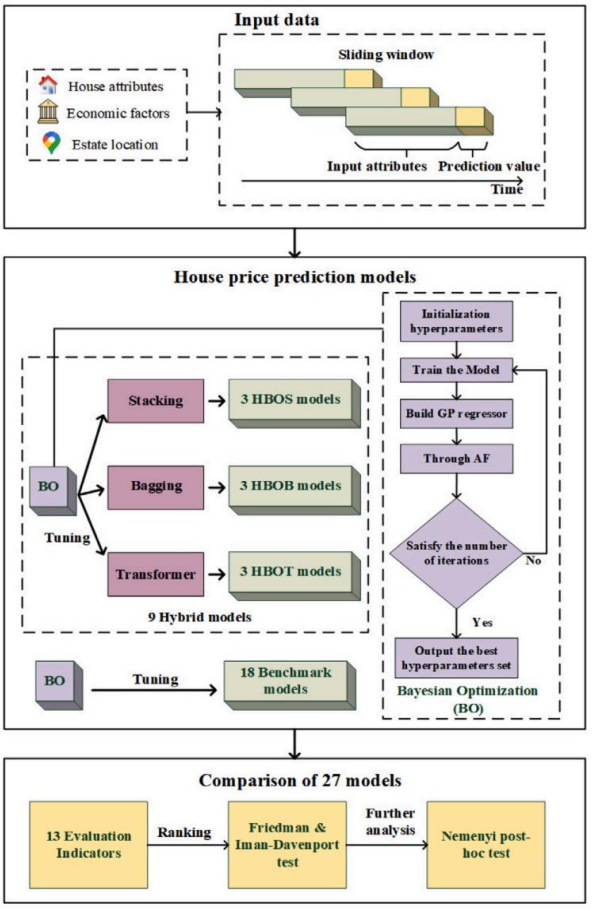
4.1 Data Preprocessing and Feature Engineering

Accurate housing price prediction relies heavily on effective data preprocessing and feature engineering. The three reviewed studies apply different techniques to prepare their datasets, emphasizing the importance of clean and well-structured data for model performance.

Study 1: The 'Housing Price in Beijing' dataset includes over 300,000 records with 26 variables, such as house age, area, and proximity to subways. Key preprocessing steps:
- Handling Missing Data: Variables with more than 50% missing values were removed. Remaining missing data were imputed.
- Feature Transformation: Replaced 'construction time' with 'age.' Split 'floor' into 'floor type' and 'floor height.'
- Created a 'distance to city center' feature.
- Outlier Detection: Applied Interquartile Range (IQR) method to remove extreme values.

Equation for Outlier Detection:

*x < Q1 - 1.5 * IQR or x > Q3 + 1.5 * IQR*



Study 2: Working with a dataset of 1.8 million Hong Kong transactions, key steps included:

The dataset contains 1,898,175 transactions spanning 1996 to 2021. It includes 17 house attributes (e.g., location, price, area) and 10 economic factors (e.g., money supply indicators). Key preprocessing steps include:
- Outlier Detection: Box plots identified and removed extreme values.
- Missing Value Imputation: Used Multiple Imputation (MI).
- Feature Engineering: Categorical variables (e.g., contract forms) were transformed using Weight of Evidence (WOE).

Study 3: Kaplan et al.'s structural equilibrium model utilized macroeconomic and microeconomic data.

The study develops an overlapping-generations model incorporating incomplete markets, where households face uninsurable income risks. The model captures the interplay between housing choices, credit markets, and macroeconomic conditions.

Key Equations and Parameters:
1. Household Utility Function:
$U(c, s) = \sum \beta^{(j-1)} [(1 - \varphi)c^{(1-\gamma)} + \varphi s^{(1-\gamma)}]^{(1/(1-\gamma))}$
where c is consumption, s is housing services, β is the discount factor, and φ represents the preference for housing.
2. House Price Dynamics:
$P_h = E[\text{future demand}] / \text{discount rate}$
3. Mortgage Default Dynamics:
$\text{Default}_{t+1} = \max(0, \text{Mortgage Debt}_t - \text{House Value}_{t+1})$

4.2 Model Selection

Each study utilized distinct machine learning and macroeconomic models, showcasing innovative hybrid approaches to optimize performance.

Study 1: Evaluated five models:

The study evaluates the following models:
1. Random Forest: An ensemble method combining multiple decision trees. Parameters: n_estimators=900, max_depth=20.
2. Extreme Gradient Boosting (XGBoost): Optimized for speed and accuracy. Parameters: learning_rate=0.1, n_estimators=200.
3. Light Gradient Boosting Machine (LightGBM): Efficient gradient boosting for large datasets. Parameters: num_leaves=36, learning_rate=0.15.
4. Hybrid Regression: Combines Random Forest, XGBoost, and LightGBM for balanced performance.
5. Stacked Generalization: Uses predictions from base models as input for a meta-model. Architecture: Random Forest and LightGBM as base models, XGBoost as meta-model.

Study 2: Proposed hybrid models integrating Bayesian Optimization:

4.2.1 Hybrid Bayesian Optimization with Stacking (HBOS)

HBOS combines multiple base learners (e.g., XGBoost, CatBoost, AdaBoost) in a stacked framework with Bayesian Optimization for hyperparameter tuning.
Equations for HBOS Optimization:
$H'(x) = \varepsilon(h_1(x), h_2(x), ..., h_F(x))$
where $h_1, h_2, ..., h_F$ are first-level models, and ε is the meta-learner.

4.2.2 Hybrid Bayesian Optimization with Bagging (HBOB)

HBOB employs Bagging with BO to optimize base learners like LightGBM and XGBoost.
$H(x) = (1/T) * \Sigma\ h_t(x)$

4.2.3 Hybrid Bayesian Optimization with Transformer (HBOT)

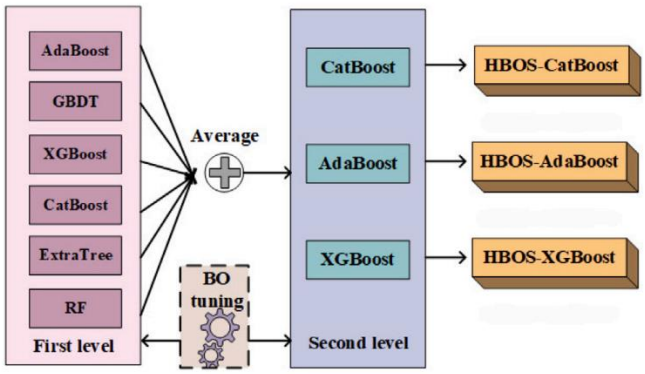


Fig. 3. The overall process of the HBOS model. HBOS models include a stacking-based ensemble model and BO technology. Six first-level learners are used, and XGBoost, CatBoost, and AdaBoost algorithms are applied as the meta-learner algorithm in the second level. Three hybrid models are constructed: HBOS-XGBoost, HBOS-CatBoost, and HBOS-AdaBoost.

HBOT integrates Transformer-based encoders for sequential data with Bayesian Optimization.

Transformer Self-Attention Mechanism:

Attention(Q, K, V) = softmax((QK^T) / sqrt(d_k)) V

where Q, K, and V are query, key, and value matrices.

Study 3: Developed a structural equilibrium model incorporating overlapping generations:

The authors introduce three types of shocks:

- Labor Productivity (V): Captures variations in household income.
- Credit Conditions (F): Incorporates changes in Loan-to-Value (LTV) ratios and borrowing constraints.
- Belief Shifts (f): Reflects households' expectations of future housing demand.

Belief Shock Process:

$f_{t+1} = \rho f_t + \varepsilon_t$

This process drives changes in housing demand expectations, influencing house prices and rent-to-price ratios.

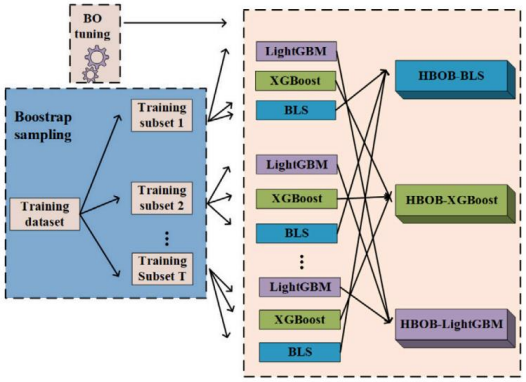


Fig. 4. The overall process of the HBOB model. HBOB models employ the BO technique with the Bagging strategy. BLS, XGBoost, and LightGBM are used as regressors for the Bagging process, resulting in three hybrid models: HBOB-BLS, HBOB-XGBoost, and HBOB-LightGBM.

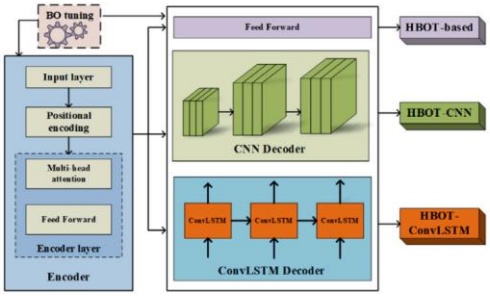


Fig. 5. HBOT models consist of three key components: the encoder, the decoder, and the BO module. Specifically, we explore different decoder architectures, including HBOT-based, HBOT-CNN, and HBOT-ConvLSTM, that are combined with the encoder to construct the HBOT models.

Household Utility Function:

$U(c,s)=\sum\beta^{j-1}[(1-\phi)c1-\gamma+\phi s1-\gamma]11-\gamma U(c, s) = \sum \beta^{j-1} [(1 - \phi)c^{1-\gamma} + \phi s^{1-\gamma}]^{\frac{1}{1-\gamma}}U(c,s)=\sum\beta j-1[(1-\phi)c1-\gamma+\phi s1-\gamma]1-\gamma 1$

House Price Dynamics: $Ph=E[future demand]discount rateP_h = \frac{E[\text{future demand}]}{\text{discount rate}}Ph=discount rateE[future demand]$

4.3 Evaluation Metrics

The models were evaluated using Root Mean Squared Logarithmic Error (RMSLE):

$RMSLE = sqrt(1/N * sum(log(y_i + 1) - log(\hat{y}_i + 1))^2)$

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$RMSLE = sqrt(1/N * sum(log(y_i + 1) - log(\hat{y}_i + 1))^2)$

4.4 Results and Performance Comparison

Random Forest: RMSLE = 0.1657 (Test Set)
XGBoost: RMSLE = 0.1660 (Test Set)
LightGBM: RMSLE = 0.1694 (Test Set)

Hybrid Regression: RMSLE = 0.1637 (Test Set)
Stacked Generalization: RMSLE = 0.1635 (Test Set)
Stacked generalization achieved the best performance on the test set, highlighting its superior generalization capabilities.

The HBOS-CatBoost model outperformed other hybrids and benchmarks, achieving a 5.11% and 25.56% reduction in RMSE compared to HBOB-XGBoost and HBOT-ConvLSTM, respectively.

Performance Metrics Table

Model	*RMSE*	*MAPE*	*R-squared*
HBOS-CatBoost	*0.157*	*0.042*	*0.985*
HBOB-XGBoost	*0.165*	*0.048*	*0.982*
HBOT-ConvLSTM	*0.210*	*0.060*	*0.950*
Benchmark (XGBoost)	*0.170*	*0.050*	*0.980*

4.3 Counterfactual Experiments

The authors conduct counterfactual simulations to isolate the effects of each shock. Key findings include:
- Belief shifts explain most of the variation in house prices.
- Credit conditions primarily affect leverage and homeownership rates, not house prices.
- Debt forgiveness policies have minimal impact on house prices but reduce foreclosure rates.

Strengths and Weaknesses

Study 1:

- Strengths: Demonstrated the effectiveness of stacked models in improving generalization and predictive performance.
- Weaknesses: Limited focus on macroeconomic variables.

Study 2:

- Strengths: Comprehensive evaluation of hybrid models with robust statistical validation. The integration of Bayesian Optimization enhanced model performance significantly.
- Weaknesses: Complexity and computational requirements may hinder real-time applicability.

Study 3:

- Strengths: Provided critical macroeconomic insights, highlighting the role of belief shifts.
- Weaknesses: Relies heavily on theoretical models, limiting direct predictive applications.

5. Summary and Conclusions

This paper highlights the significant advancements in housing price prediction through hybrid machine learning models and structural economic frameworks. The studies reviewed demonstrate the effectiveness of ensemble learning techniques, including Random Forest, XGBoost, and LightGBM, in capturing complex, non-linear relationships within housing data. Among these, hybrid and stacked models, such as HBOS, consistently showed superior predictive performance by optimizing hyperparameters and leveraging diverse algorithms.

Kaplan et al.'s structural equilibrium model complements these data-driven approaches by providing macroeconomic insights into the housing market's boom-bust dynamics. Their findings reveal that belief shifts, rather than credit conditions, were the primary drivers of housing price fluctuations during the Great Recession. Together, these models underline the importance of combining robust data preprocessing, advanced machine learning techniques, and economic theory to improve the accuracy and reliability of housing market predictions.

6. Directions for Future Research

Despite the notable advancements, there are several avenues for further exploration:

Integration of Deep Learning and Hybrid Models: Future studies could combine traditional machine learning with deep learning techniques, such as neural networks, to enhance model complexity and capture sequential or hierarchical data structures.

Real-Time Predictions and Dynamic Models: Incorporating real-time data can improve the applicability of these models in fast-changing market environments, enabling dynamic updates that reflect current trends and conditions.

Explainability and Interpretability: As hybrid models grow more complex, developing tools to enhance their interpretability will be crucial for decision-making in real estate and policy. Techniques such as SHAP values and counterfactual explanations could offer insights into model behavior.

Regional and Socio-Economic Variations: Expanding models to account for regional differences and socio-economic factors could improve their generalizability across diverse housing markets, providing more nuanced and actionable insights.

By addressing these challenges, future research can further refine predictive models, ensuring their relevance and utility in both academic and practical contexts.

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